Fall’itis Engineering Journal

Bio Page

Fall'itis is a team comprising two members: Michele Tesser and Alin Odobescu. Both members have recently completed their fifth year of high school, specializing in computer science. We are also enrolled in a University's computer science program, starting from the next academic year.

Mentor

Fabrizio Vendramin

We consider ourselves fortunate to have Fabrizio Vendramin as our mentor. With his extensive experience in the RoboCup, Fabrizio has provided us with invaluable guidance and advice. As the first group from our school to participate in a simulated robotics competition, we greatly rely on Fabrizio's expertise to support our journey.

Members

Alin Odobescu

Alin Odobescu's passion for computer science has flourished since middle school. He has actively participated in competitive programming contests in recent years. While his interest in the robotics field is relatively new, Alin has already made significant contributions in developing the mapping system, path planning, and following aspects. His involvement in this competition has ignited a keen interest in machine learning, and robotics.

Michele Tesser

Michele Tesser assumed responsibility for object detection, dataset management, operating scoring game elements, and meticulously analyzing various corner cases. His passion for mathematics has been instrumental in driving the competition's progress.

Other Greetings

Diego Chinellato

The team wishes to extend their heartfelt gratitude to Professor Diego Chinellato for his invaluable support in the domains of mathematics and computer vision. Diego's guidance and insightful contributions have been immensely beneficial to our entire journey.

Awards

Robotics

Fall'itis emerged victorious in the national phase of the RoboCup Junior Rescue Simulation category, securing their rightful place in the world finals.

Other Fields

Michele has achieved Italian excellence in statistics competitions. Meanwhile, Alin has attained the national level in computer science competitions. Both members have actively participated in mathematics and computer science competitions prior to this event, fostering a strong synergy and deepening their mutual understanding.

Operational logistics

Meetings

The advantage of being in the same class allowed us to engage in daily face-to-face discussions, ensuring a consistent flow of feedback among ourselves. We had the privilege of meeting practically every day, leveraging the allocated laboratory hours granted by our teacher to devote ourselves to the project.

Communication

We relied on Telegram as the primary mode of communication, enabling real-time messaging and serving as a platform for storing, sharing, and accessing data and resources. Additionally, daily interactions within the classroom facilitated continuous collaboration and prompt feedback. Discord was also utilized for voice and streaming communication purposes.

Version Control System

To streamline version control and collaboration, we leveraged the power of GitHub. This platform proved indispensable in ensuring seamless synchronization among the team members, enabling us to work together with efficiency and effectiveness on our individual tasks. By harnessing the capabilities of GitHub, we maintained a synchronized workflow and effectively tracked changes made to our project.

Research

During the initial stages of the project, we heavily relied on Obsidian for the organization of our thoughts and information, encompassing mathematical formulae and conducting research. However, the Obsidian directory is currently unavailable to us.

Daily Log

Below is the daily log of the Fall'itis team, excluding days with minimal activity or small events that were carried forward to the subsequent day to enhance readability. Additionally, we have incorporated a concise summary of significant events occurring each month, these summaries were integrated into the document once it has been completed, enhancing its overall readability.

Initially, Alin was primarily responsible for the model and image recognition aspect, while Michele focused on the controller. However, as the project progressed, the roles underwent an inversion. In reality, we did not have rigidly defined roles as each team member actively contributed and collaborated on all aspects of the robot, providing support and assistance wherever needed.

January

In January, we initiated our project, acquiring valuable knowledge and diligently organizing our efforts. Notably, we established a GitHub repository, successfully implemented our initial controller, and gained proficiency in utilizing TensorFlow.

January 14, Alin, Michele

Tasks done today

- We had a meeting with our mentor, Fabrizio.
- We acquainted ourselves with the rules of the competition.
- We collected information regarding Webots and Erebus.
- We commenced studying computer vision and object detection models.
- We stumbled upon the opportunity of utilizing a lidar in the competition.

New Ideas | Thoughts

We are considering the utilization of an object detection model or image classifier to identify the signs. This approach appears favorable compared to relying solely on computer vision and defining a set of rules that could potentially falter in certain circumstances. Additionally, we are currently assessing the feasibility of incorporating a lidar system to generate a grid representation of the environment. This would enable us to perform path planning and obstacle avoidance more effectively on the said grid.

Research

Computer vision:

- http://szeliski.org/Book

Object detection:

- https://jonathan-hui.medium.com/object-detection-series-24d03a12f904
- https://blog.roboflow.com/object-detection

Webots LiDAR:

- https://cyberbotics.com/doc/reference/lidar

January 18, Alin, Michele

Tasks done today

- We continued our research on computer vision, delving deeper into the subject.
- After careful consideration, we made the decision to implement a Convolutional Neural Network (CNN) in order to develop a robust image classification model.

New Ideas | Thoughts

Regrettably, there has been limited progress in the last few days due to various commitments and obligations. However, today we have made a significant decision to employ a convolutional neural network (CNN) for the purpose of constructing an image classification model specifically tailored for the sign detection task.

Research

Computer vision:

- https://towardsdatascience.com/everything-you-ever-wanted-to-know-about-computer-vision-heres-a-look-why-it-s-so-awesome-e8a58dfb641e

Image classification using pytorch:

- https://medium.com/bitgrit-data-science-publication/building-an-image-classification-model-with-pytorch-from-scratch-f10452073212

January 20, Alin, Michele

Tasks done today

- We have allocated specific roles for each team member in the project. Michele will be responsible for navigation and mapping, while Alin will focus on sign recognition, model tuning, and data labeling.
- Furthermore, Alin has created a task list, as depicted in Figure 1, outlining various assignments.

New Ideas | Thoughts

At present, the team has not yet finalized a name. We have scheduled a discussion in the upcoming days to determine the most suitable name for our team.

Figures/Drawings/Tables

Figure 1, Alin's tasks

Tasks

- [ ] learn about webots viewport and moving around
- [ ] learn about webots robot controller
- [ ] learn about webots robot components and how to access and use them
- [ ] learn how machine learning works
- [ ] learn how object classification and identification works
- [ ] train an object classification artificial intelligence using pytorch/tensorflow or another machine learning framework
- [ ] learn about object detection
- [ ] implement an object detector using the previously trained artificial intelligence

January 22, Alin, Michele

Tasks done today

- Today marks a significant milestone as we have officially selected a team name. After careful deliberation, we are proud to announce that the name "Fall'itis" has been chosen for our team.

January 26, Alin, Michele

Tasks done today

- We have successfully established a project repository on GitHub, opting to maintain a single branch named "main" and committing directly to it due to our team's small size.
- In accordance with our pre-planned schedule, we held a meeting with Professor Diego for a brief lesson on utilizing TensorFlow, a machine learning library similar to PyTorch.
- During the meeting, we extensively discussed project organization and ultimately decided to adopt a LiDAR-based mapping approach instead of a tile-based one. This choice was primarily motivated by the requirement of developing a distinct controller for the fourth zone in the event of a tile-based mapping system.

Issues and solutions

We encountered difficulties with the map creator in Webots, as it was not loading the map correctly. However, we were able to resolve this issue by implementing a small Python script that modifies the exported map accordingly.

New Ideas | Thoughts

Currently, we are still in the process of figuring out how to effectively utilize the lidar for creating an accurate map of the environment. According to the Webots documentation we

can obtain a range scan of the lidar and directly a point cloud from Webots. However, it is worth noting that the latter may be computationally expensive.

January 28, Alin, Michele

Tasks done today

- To enhance organization within our GitHub repository, we have implemented separate folders for models, data, and a source directory housing all relevant files.
- Python has been chosen as our programming language due to its simplicity and the availability of excellent libraries such as NumPy. In order to maintain code legibility and clarity, we have opted to adhere to the PEP8 Python style guide.
- Excitingly, we have uncovered a recently introduced feature enabling the utilization of an external controller with Erebus. As a result, we have successfully configured our programming environment to accommodate this capability.

Research

PEP8 style guide:

- https://peps.python.org/pep-0008/
- https://realpython.com/python-pep8/

January 30, Alin, Michele

Tasks done today

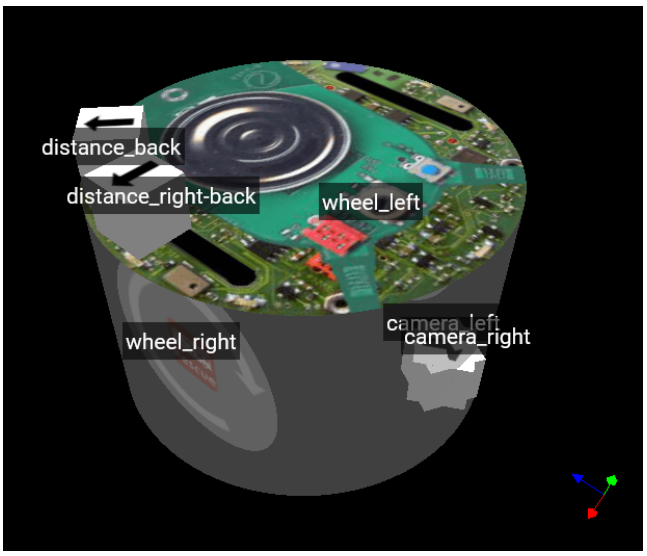
- We have successfully developed our initial controller, which involves a robot equipped with two right-facing distance sensors capable of wall-following behavior. Furthermore, we have installed two cameras on the robot, positioned in opposite directions as illustrated in Figure 1. This enables us to create a small dataset to train our model.

New Ideas | Thoughts

We are delighted with the progress we have made with our first robot, despite its current limitations. It serves as an important initial step in our journey towards developing more sophisticated and capable systems.

Figures/Drawings/Tables

Figure 1, our first robot



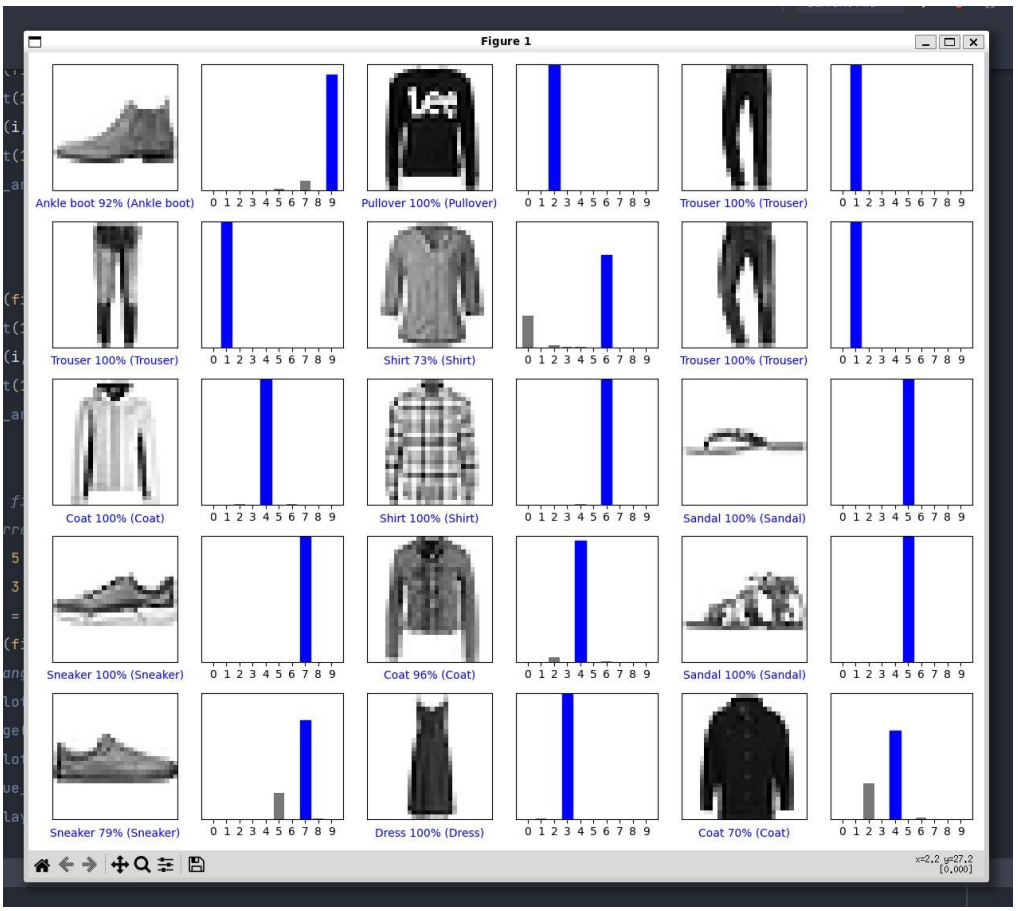
January 31, Alin, Michele

Tasks done today

- We have embarked on the journey of learning how to utilize TensorFlow for creating an image classification model, benefiting from the knowledge imparted by our professor, Diego. By applying this newly acquired knowledge, we have successfully replicated the model presented in the TensorFlow documentation. Figure 1 showcases a matplotlib plot illustrating the model's predictions.

Figures/Drawings/Tables

Figure 1, model's predictions



Research

TensorFlow documentation:

- https://www.tensorflow.org/tutorials

February

In February, we reached a noteworthy milestone by creating a refined controller that surpassed basic wall tracking. Additionally, we successfully generated a labeled dataset encompassing various sign types, facilitating the training of an image classification model using TensorFlow.

February 3, Alin, Michele

Tasks done today

- In order to assess their precision, we have experimented with the GPS and inertial unit functionalities. Based on our evaluation, we have concluded that these sensors provide sufficient accuracy for calculating the position and rotation of the robot. Consequently, we have decided to rely on them, eliminating the need for odometry derived from wheel encoders. This approach avoids the necessity of implementing SLAM techniques for environment mapping.

February 4, Alin, Michele

Tasks done today

- Recognizing the need for an organized structure in our robot controller, we conducted research on the state machine pattern.

Research

State Machines:

- https://en.m.wikipedia.org/wiki/State_pattern#:~:text=The%20state%20pattern%20is%20a,concept%20of%20finite-state%20machines
- https://en.m.wikipedia.org/wiki/Finite-state_machine

February 8, Alin, Michele

Tasks done today

- We have begun the process of expanding our dataset by collecting additional samples and labeling them accordingly.
- Using the architecture outlined in Figure 1, we proceeded to train a model on the augmented dataset. The model's outcomes are displayed in Figure 2.

Figures/Drawings/Tables

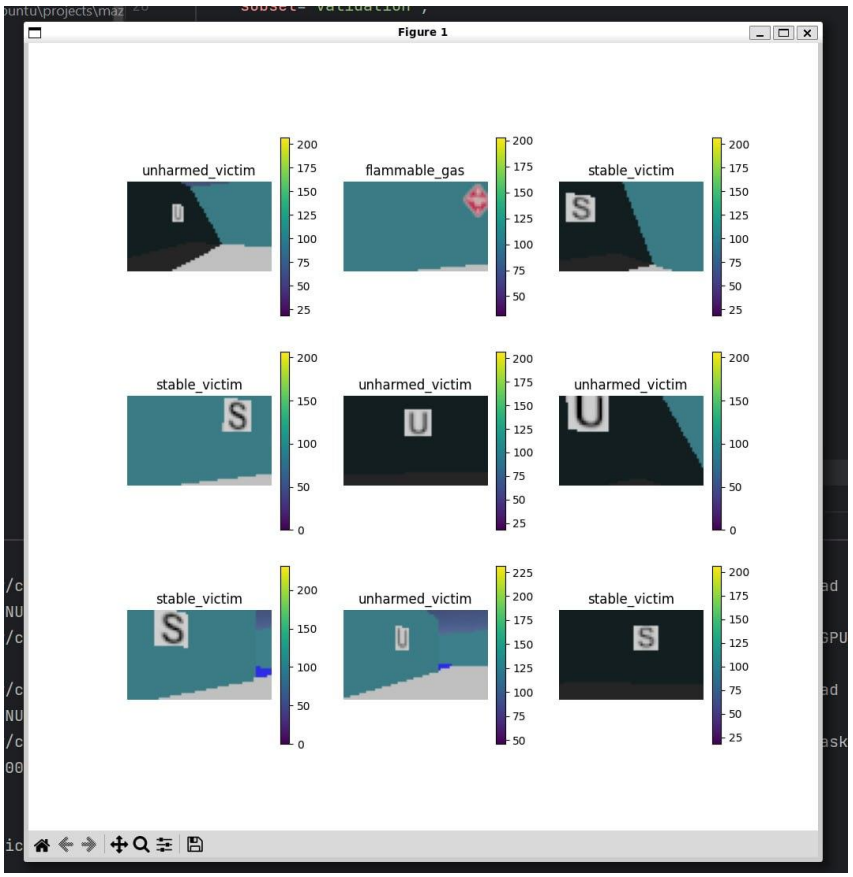
Figure 1, architecture of our model

```python

model = tf.keras.Sequential([
    tf.keras.layers.Rescaling(1. / 255),
    tf.keras.layers.Conv2D(32, 3, activation='relu'),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Conv2D(32, 3, activation='relu'),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Conv2D(32, 3, activation='relu'),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(num_classes)
])

```

Figure 2, model’s predictions



February 11, Alin, Michele

Tasks done today

- We have successfully developed a robot controller, which possesses the capability to navigate by moving to specified positions and rotating to desired orientations. This controller marks an important milestone in our project's progress. Video 1 is a showcase of our current controller.

Issues and solutions

During our observations, we have detected certain irregularities between tiles of different colors.

Figures/Drawings/Tables

Video 1, showcase of our controller
https://drive.google.com/file/d/1MaSp9DONlSea2sEWpDvkT-D13_raPTnO/view

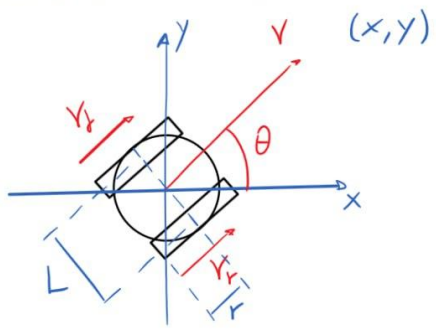
February 15, Alin, Michele

Tasks done today

- We have enhanced the controller's performance by implementing the formulas illustrated in Figure 1. These formulas provide guidance on effectively maneuvering a differential wheeled robot, leading to improved control and navigation.

Figures/Drawings/Tables

Figure 1, kinematics of a differential wheeled robot



$$\theta_d = \theta_t - \theta$$

$$\theta_{er} = \operatorname{atan2}(\sin(\theta_d), \cos(\theta_d))$$

$$\dot{x} = v\cos(\theta)$$
$$\dot{y} = v\sin(\theta)$$
$$\dot{\theta} = \omega = \frac{(v_r - v_l)}{L}$$
$$v = \frac{(v_l + v_r)}{2}$$

$$v = \sqrt{\dot{x}^2 + \dot{y}^2}$$
$$v_r = \left(v + \frac{L}{2}\omega\right)$$
$$v_l = \left(v - \frac{L}{2}\omega\right)$$

Research

Differential wheeled robot:

- https://en.wikipedia.org/wiki/Differential_wheeled_robot

February 19, Alin, Michele

Tasks done today

- Following our discussions on code organization, we have reached a consensus on the proposed division of the controller into a Robot class and an Engine class, as illustrated in Figure 1. The Robot class handles the higher-level logic and behavior of the robot, while the Engine class is responsible for driving the robot by executing the necessary commands.
- We have generated a plot, as depicted in Figure 2, showcasing the accuracy of our trained model throughout different epochs on both the training set and validation set. This visualization provides insights into the model's performance during the training process.

Issues and solutions

Upon analyzing the plot in Figure 2, we have observed some signs of overfitting in our model. To address this issue, we have planned to dedicate the upcoming days to enhancing both the architecture of our model and the dataset used for training.

Figures/Drawings/Tables

Figure 1, current controller implementation

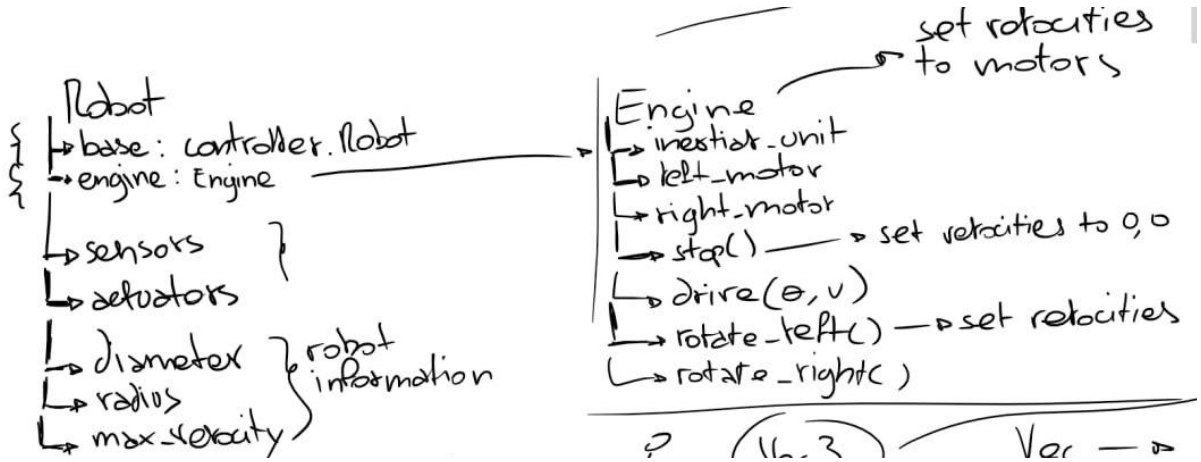
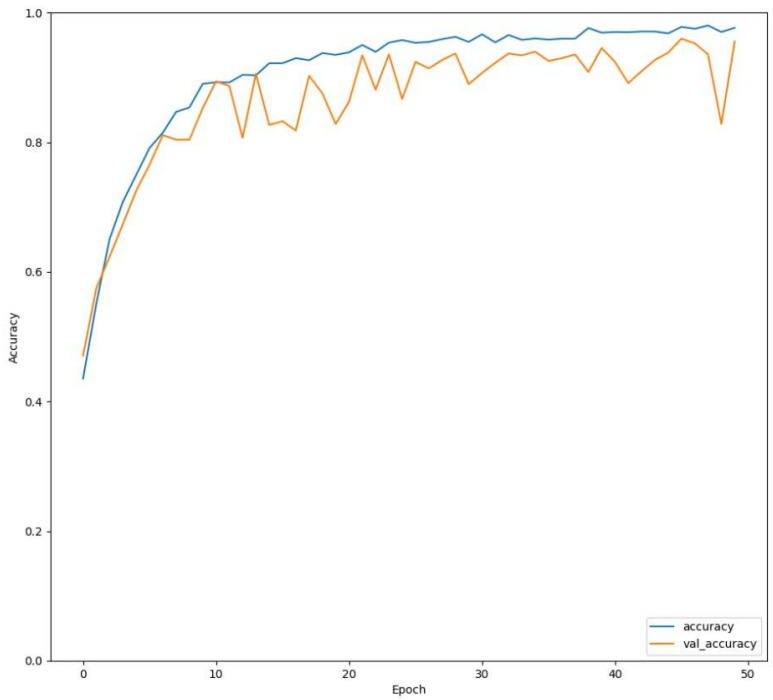


Figure 2, accuracy of our model on training and validation sets



February 27, Alin, Michele

Tasks done today

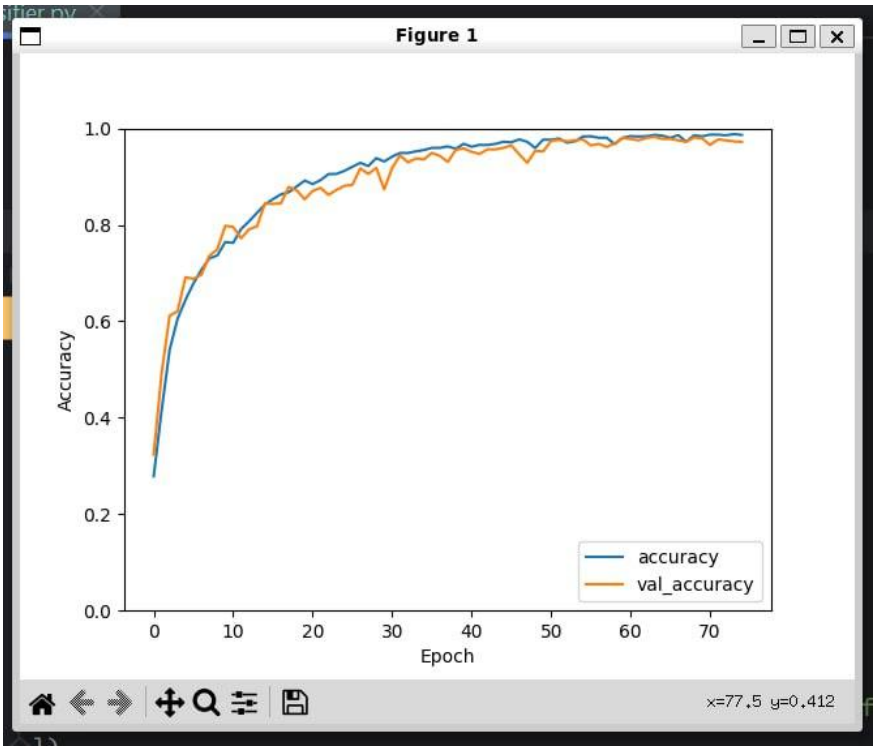
- We have successfully enhanced the accuracy of our model by making improvements to our dataset, as illustrated in Figure 1. These enhancements have resulted in improved training and validation performance, ultimately leading to a more accurate and reliable model.

New Ideas | Thoughts

In the upcoming days, we are planning to initiate the implementation of the mapping system using the lidar sensor. Given that we have a functional controller and a trained model, we are well-equipped to proceed with this task.

Figures/Drawings/Tables

Figure 1, accuracy of our model on training and validation sets



March

In March, significant progress was made, including the initial mapping of the environment using lidar, initial path planning, and sign detection.

March 6, Alin, Michele

Tasks done today

- Through the utilization of Webots' lidar capabilities, we successfully obtained a point cloud dataset. We then employed Matplotlib to visualize and plot the acquired lidar points, as demonstrated in Figure 1.

New Ideas | Thoughts

Upon careful observation, we recognized that the computational cost associated with calculating the lidar point cloud using Webots' feature was substantial. Consequently, we made the decision to perform scans at intervals, striking a balance to avoid excessive code execution slowdown.

Figures/Drawings/Tables

Figure 1, visualization of the filtered lidar-derived point cloud data

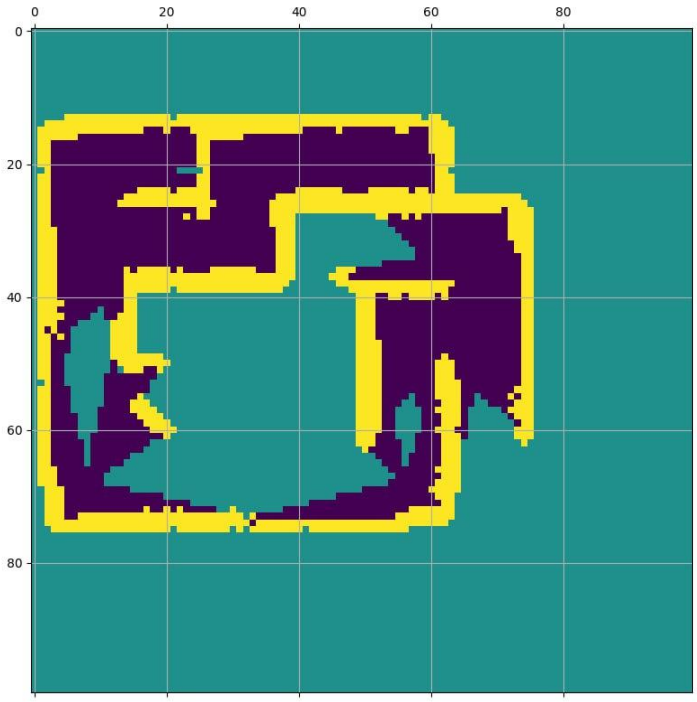
March 10, Alin, Michele

Tasks done today

- By converting the lidar-derived point cloud into a grid format, we applied the Bresenham line algorithm to differentiate between free and unknown points on the grid. The resulting plot, depicted in Figure 1, showcases the outcome of this process.

Figures/Drawings/Tables

Figure 1, plot illustrating the occupancy grid generated using the lidar-derived point cloud



Research

Occupancy grid mapping:

- https://atsushisakai.github.io/PythonRobotics/modules/mapping/lidar_to_grid_map_tutorial/lidar_to_grid_map_tutorial.html
- https://en.wikipedia.org/wiki/Occupancy_grid_mapping
- https://en.wikipedia.org/wiki/Bresenham%27s_line_algorithm

March 15, Alin, Michele

Tasks done today

- In an effort to enhance efficiency and gain greater control over calculations, we opted for an alternative approach utilizing the range image returned by the lidar instead of directly obtaining the point cloud in global coordinates from Webots. Figure 2 exhibits the results achieved through our manual calculation of the point cloud.

Issues and solutions

During our implementation, we encountered challenges with accurately positioning the lidar scans on the grid, as shown in figure 1. However, we successfully resolved this issue by revising the method used to convert world coordinates into grid coordinates.

Figures/Drawings/Tables

Figure 1, the issue we encountered with coordinate conversion after three lidar scans

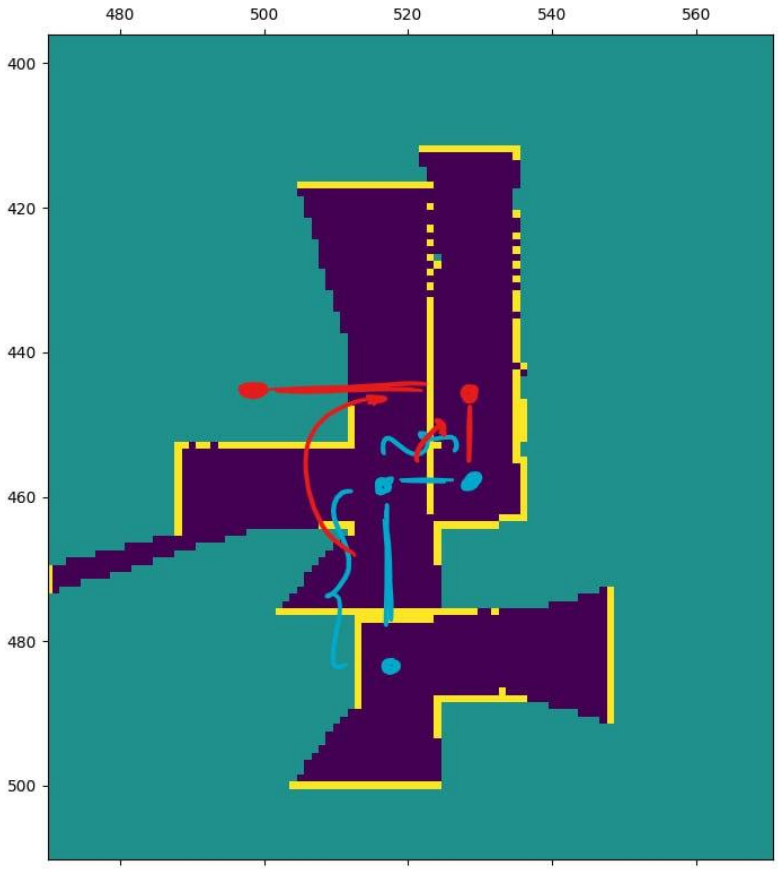
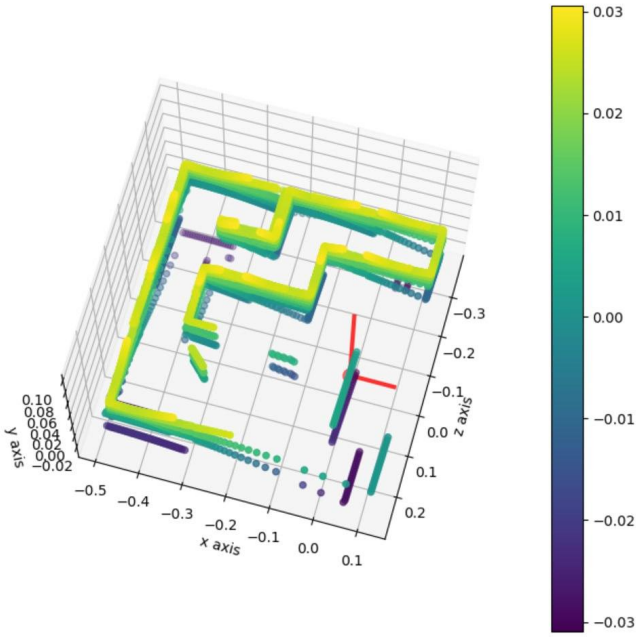


Figure 2, 3D plot depicting the manually calculated point cloud obtained from the range image



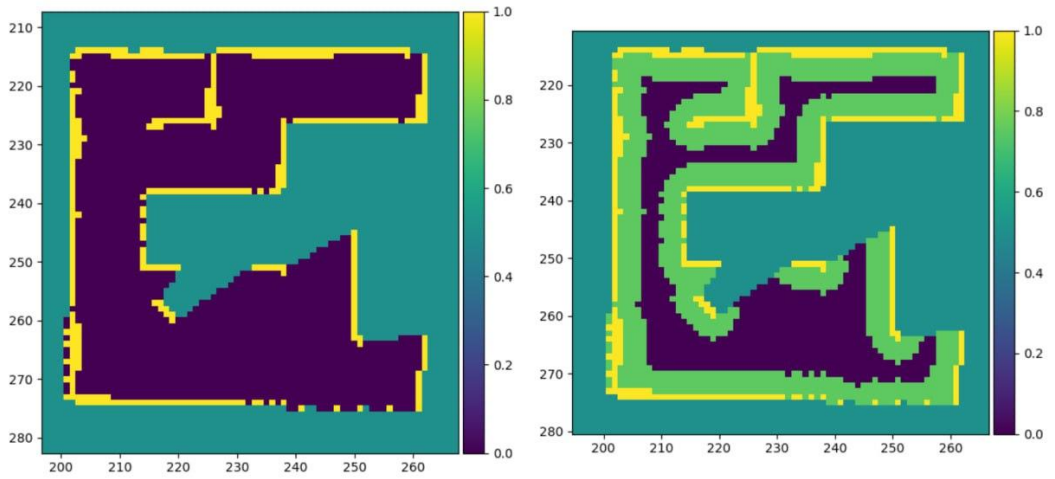
March 20, Alin, Michele

Tasks done today

- We have initiated the development of the path-finding algorithm. To ensure safety from obstacles, we implemented a binary dilation operation on the grid. This expansion of the walls enables us to calculate a secure path for the robot to navigate.

Figures/Drawings/Tables

Figure 1, the occupancy grid and the same grid with a dilation morphological transformation applied



March 21, Alin, Michele

Tasks done today

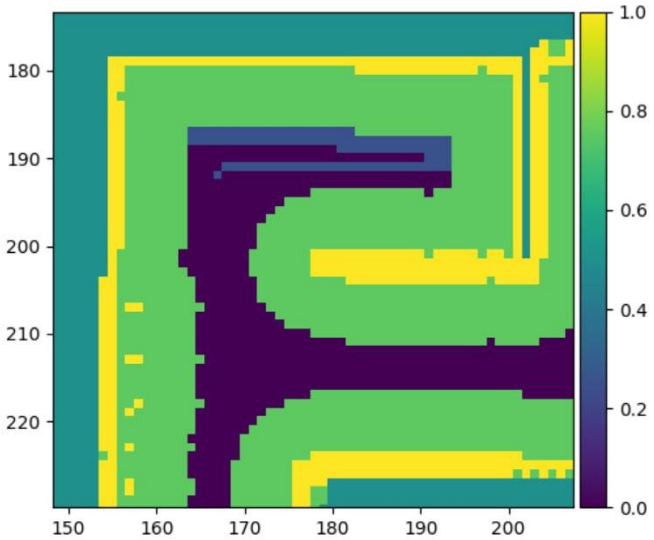
- Our objective is to enable the robot to detect signs while following the walls closely. To achieve this, we have devised a pathfinding algorithm that keeps the robot in proximity to the dilated area of the grid.

Issues and solutions

As depicted in Figure 1, our algorithm has revealed certain issues that we intend to address and rectify in the upcoming days.

Figures/Drawings/Tables

Figure 1, the path obtained from our pathfinding algorithm



March 22, Alin, Michele

Tasks done today

- We have successfully resolved the issue with our path planning algorithm that arose yesterday. The problem was rooted in flawed logic, which has now been rectified. In order to function correctly, the path planning algorithm now maintains the stored direction, as shown in Figure 1. Figure 2 depicts the path planned by the algorithm, effectively following the dilated area as intended.
- To determine the closest dilated area from the current position, we employed a breadth-first search algorithm. This algorithm provides both the nearest dilated area and the corresponding direction for the wall-following algorithm.

Figures/Drawings/Tables

Figure 1, schematic illustrating the behavior of our algorithm based on the given starting direction

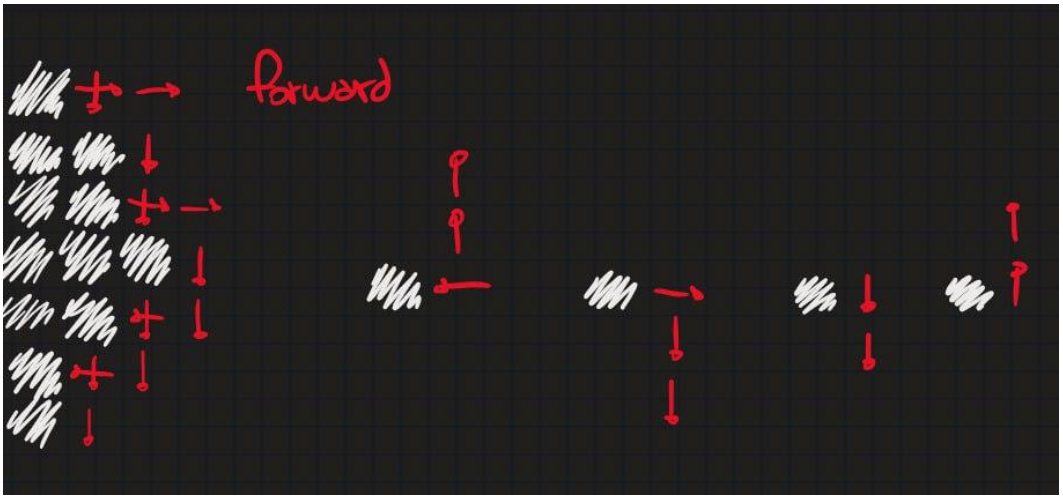
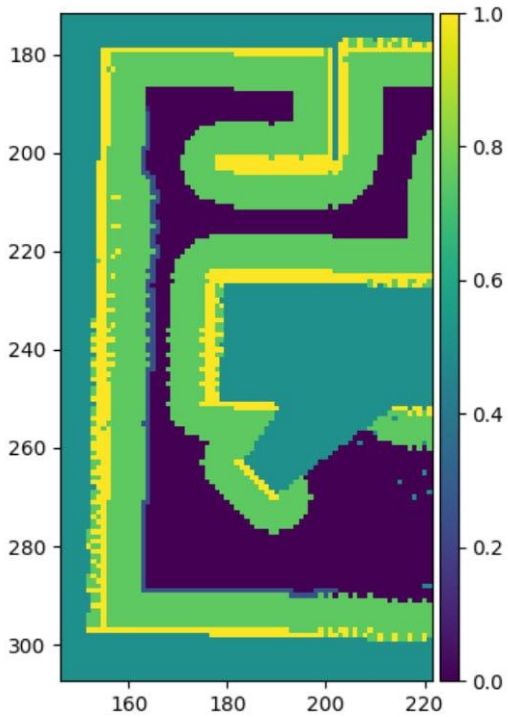


Figure 2, the path devised through the implementation of our refined algorithm



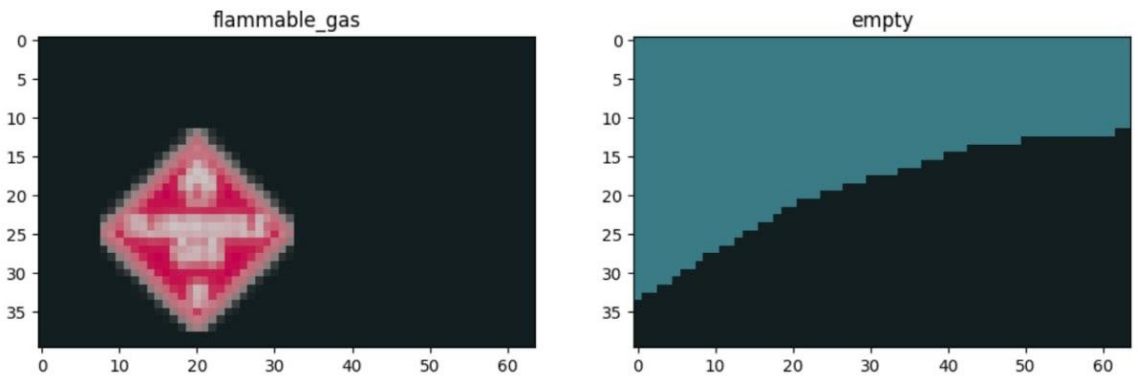
March 25, Alin, Michele

Tasks done today

- We utilized our pre-trained image classification model to detect signs on the map. Figure 1 displays a plot containing images captured by the two cameras currently in use.

Figures/Drawings/Tables

Figure 1, plot showcasing the camera images alongside their corresponding detected classifications, as provided by the image classification model



April

In April, we made significant advancements in our mapping algorithm, enhancing both its accuracy and efficiency. We have also made notable improvements to our path planning algorithm, ensuring thorough exploration of all regions within the maze.

April 4, Alin, Michele

Tasks done today

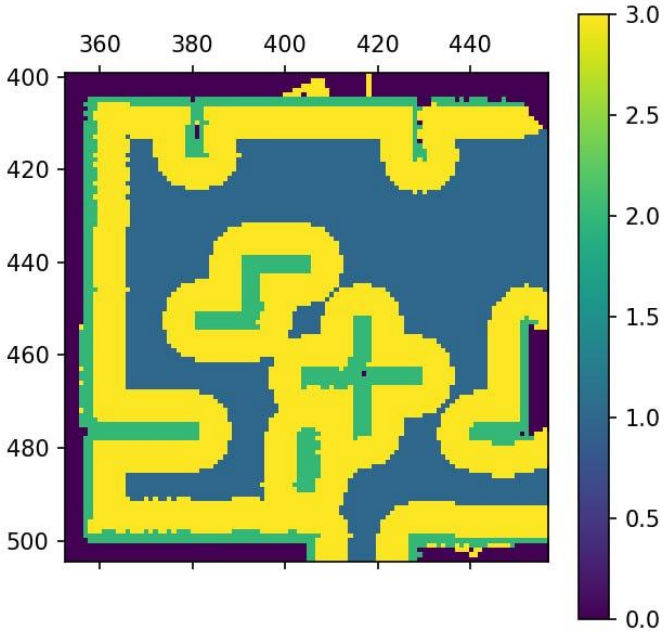
- We retrieved the maps from last year's competition and conducted thorough testing of the robot on those maps.

Issues and solutions

During testing on the ninth map of the 2022 competition, we encountered an issue with our robot's ability to navigate through narrow spaces, as depicted in figure 1.

Figures/Drawings/Tables

Figure 1, the dilation of the map encroaches upon walkable space within narrow passages



April 6, Alin, Michele

Tasks done today

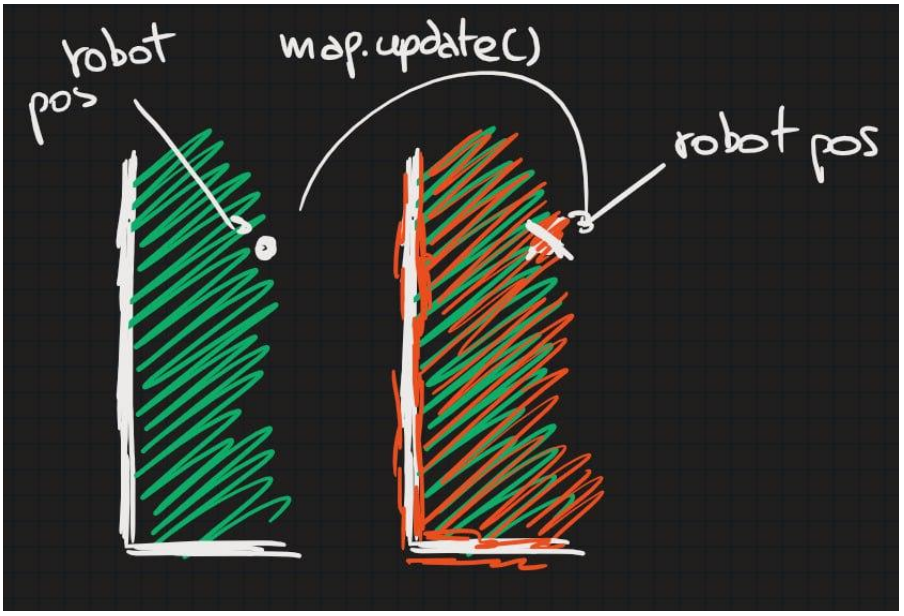
- Today, we proceeded with further testing and experimentation with the robot.

Issues and solutions

During our ongoing testing, we encountered another intermittent issue, as depicted in Figure 1. Our algorithm occasionally faced path disruptions when a previously empty space was marked as a wall dilation in subsequent updates. To address this, we implemented a check during each update to ensure path validity. If the path was found to be invalid, we utilized a breadth-first search to restore the path to the nearest valid location.

Figures/Drawings/Tables

Figure 1, a schematic illustrating the problem we encountered with path planning during map updates



April 10, Alin, Michele

Tasks done today

- We have significantly enhanced the efficiency of our mapping algorithm. Rather than calculating the position for every point in the lidar point cloud, we utilized the range image provided by the lidar to generate a polar grid. This polar grid was then efficiently converted to Cartesian coordinates using pre-calculated indices with the help of numpy. Figure 1 displays the polar grid, while Figure 2 showcases the corresponding polar grid transformed into Cartesian coordinates.

New Ideas | Thoughts

Thanks to the improved efficiency of our mapping algorithm, we have achieved the capability to update the map during every Webots step. This significant enhancement allows us to maintain an up-to-date and accurate representation of the environment in real-time.

Figures/Drawings/Tables

Figure 1, polar grid representation of the lidar scan

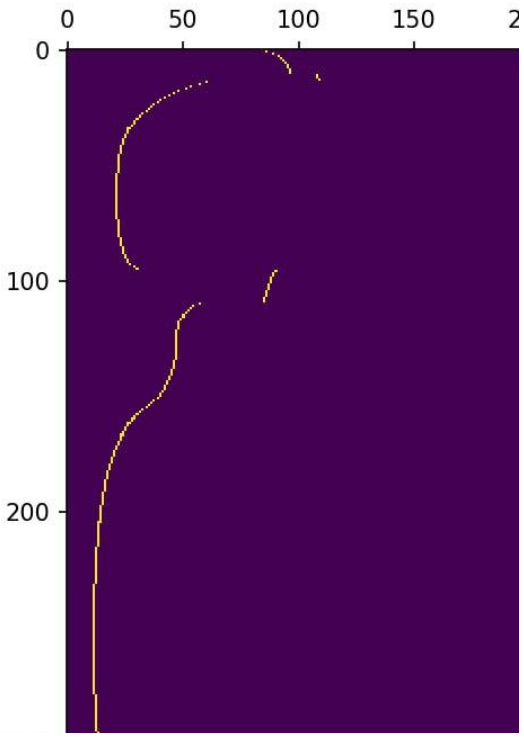
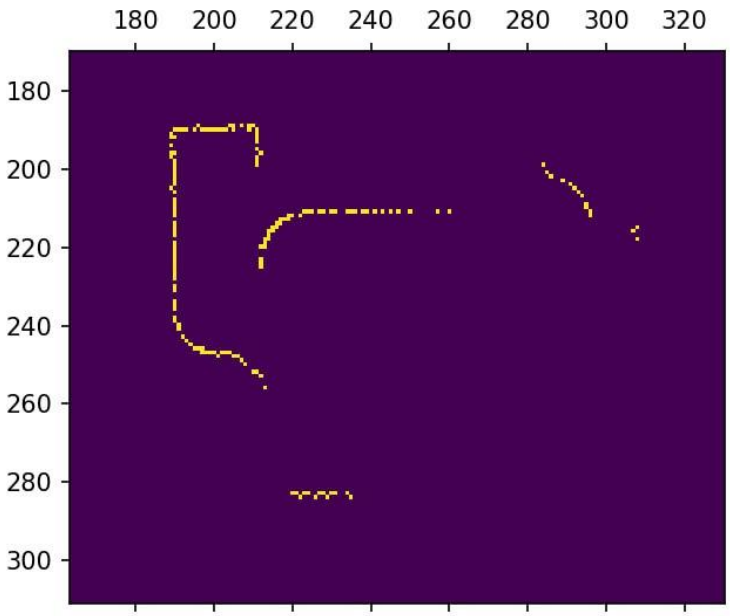


Figure 2, Cartesian grid obtained by converting the polar grid



April 13, Alin, Michele

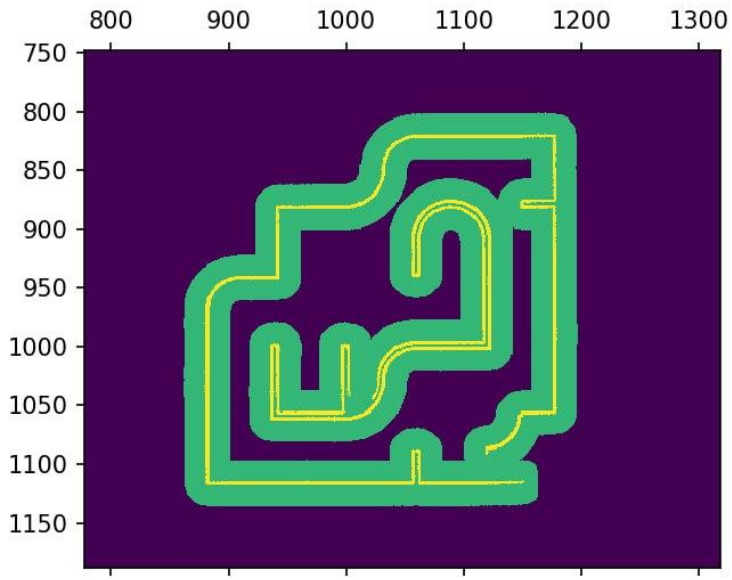
Tasks done today

- To resolve the issue encountered on April 4th, we implemented two solutions. First, we slightly reduced the dilation radius, allowing the robot to navigate without getting blocked by obstacles. Additionally, we increased the resolution of the occupancy grid, providing a finer level of detail for better path planning and obstacle avoidance.

Figure 1 showcases the updated occupancy grid mapping, reflecting these improvements.

Figures/Drawings/Tables

Figure 1, plot of the occupancy grid showcasing the recent improvements that have been implemented



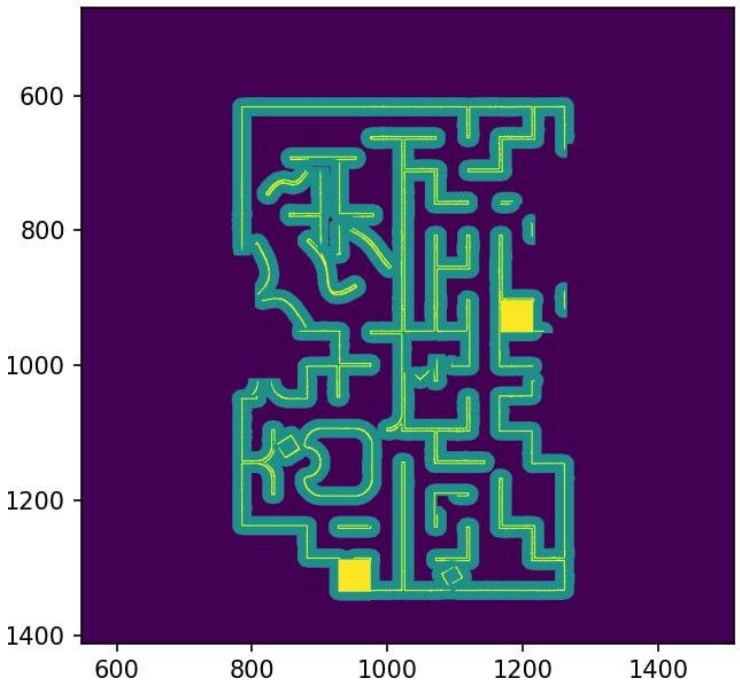
April 16, Alin, Michele

Tasks done today

- To address the challenge of detecting holes, we incorporated three front-facing distance sensors on the robot. This setup allows us to detect holes even when entering a tile at a narrow angle. We successfully mapped the detected holes onto the occupancy grid, as demonstrated in Figure 1.

Figures/Drawings/Tables

Figure 1, plot of the occupancy grid, revealing two noticeable holes indicated by the large yellow squares



April 19, Alin, Michele

Tasks done today

- We made an attempt to detect signs using OpenCV and image processing techniques. Figure 1 displays the results obtained using color segmentation, while Figure 2 shows the outcomes achieved by highlighting the edges in the image.

New Ideas | Thoughts

We were unable to detect all the signs in the small amount of time we allocated for the attempt. As a result, we have made the decision to continue relying on the model, which has proven to perform better across a wider range of circumstances.

Figures/Drawings/Tables

Figure 1, color segmentation applied to an image of a flammable gas

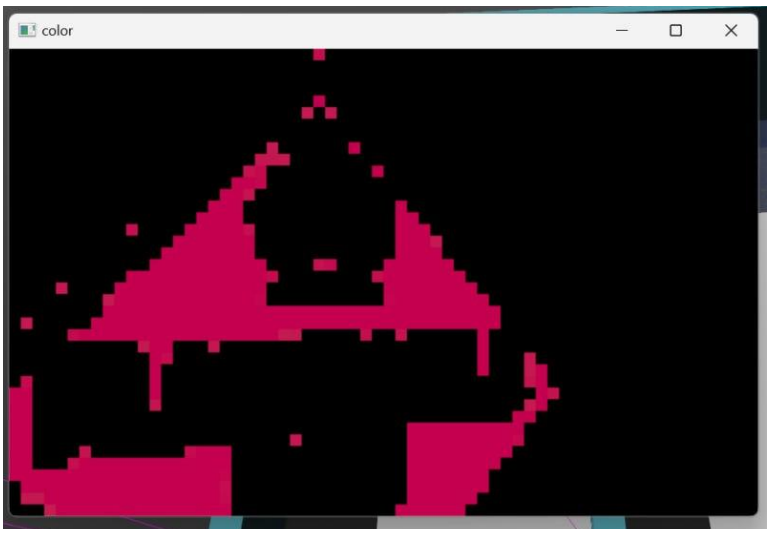
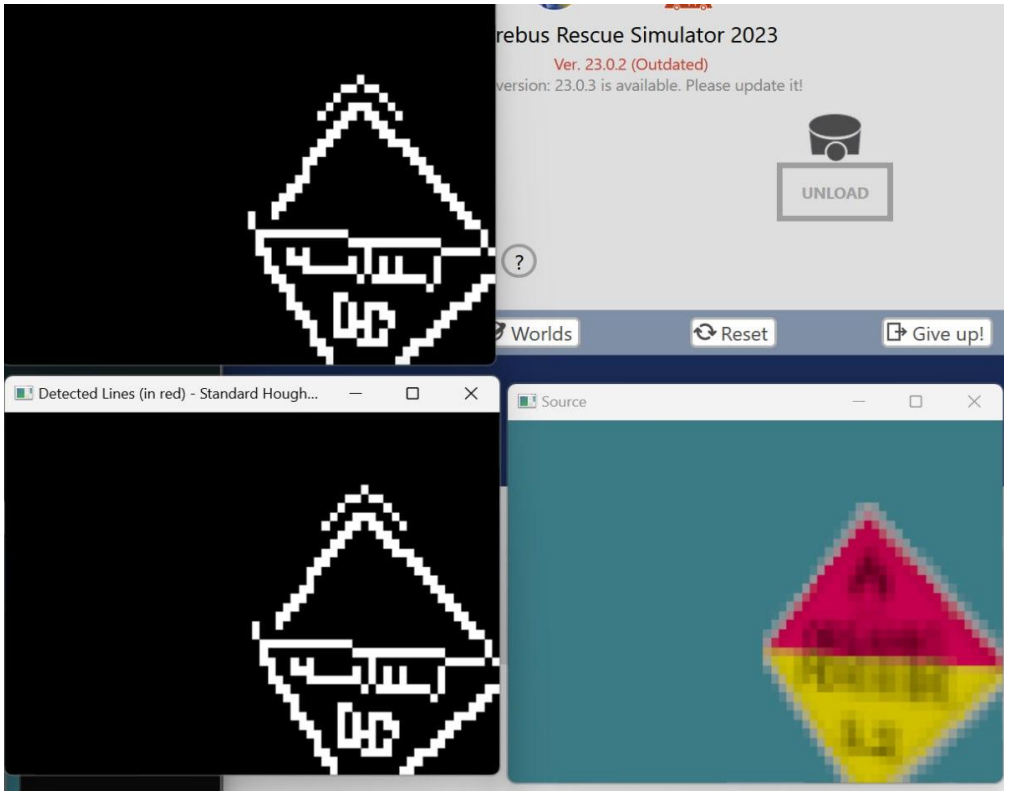


Figure 2, edge detection applied to an image of a hazmat sign



Research

Color segmentation:

- https://stackoverflow.com/questions/47483951/how-to-define-a-threshold-value-to-detect-only-green-colour-objects-in-an-image/47483966#47483966
- https://stackoverflow.com/questions/51229126/how-to-find-the-red-color-regions-using-opencv/51230291#51230291
- https://stackoverflow.com/questions/10948589/choosing-the-correct-upper-and-lower-hsv-boundaries-for-color-detection-withcv/48367205#48367205

Detecting edges:

- https://docs.opencv.org/3.4/d9/db0/tutorial_hough_lines.html

- https://docs.opencv.org/3.4/da/d5c/tutorial_canny_detector.html

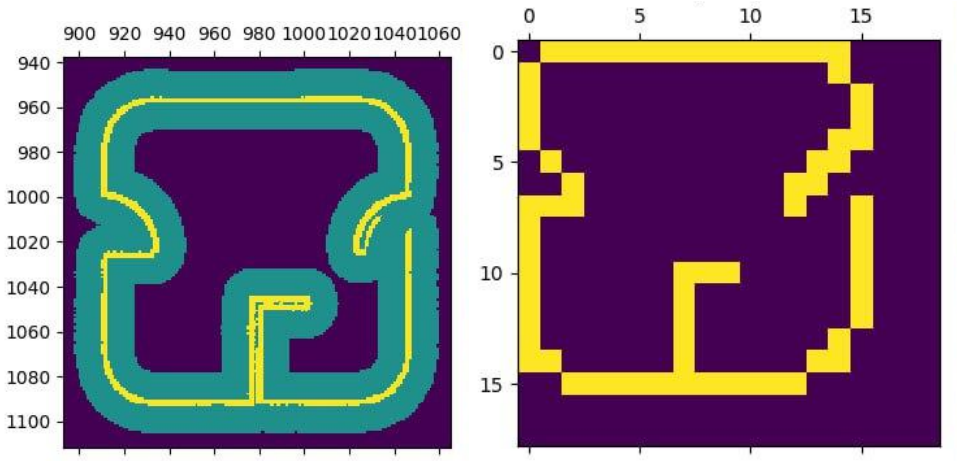
April 21, Alin, Michele

Tasks done today

- We have initiated the process of transforming our occupancy grid into the grid format required by Erebus for the mapping bonus. Figure 1 represents our initial attempt at converting the walls to the mapping bonus grid. To accomplish this, we divided our map into large regions and determined the presence of walls based on a specified threshold. If a region exceeded the threshold, the corresponding cell in the bonus grid was designated as a wall.

Figures/Drawings/Tables

Figure 1, the occupancy grid on the left and the corresponding map bonus grid on the right



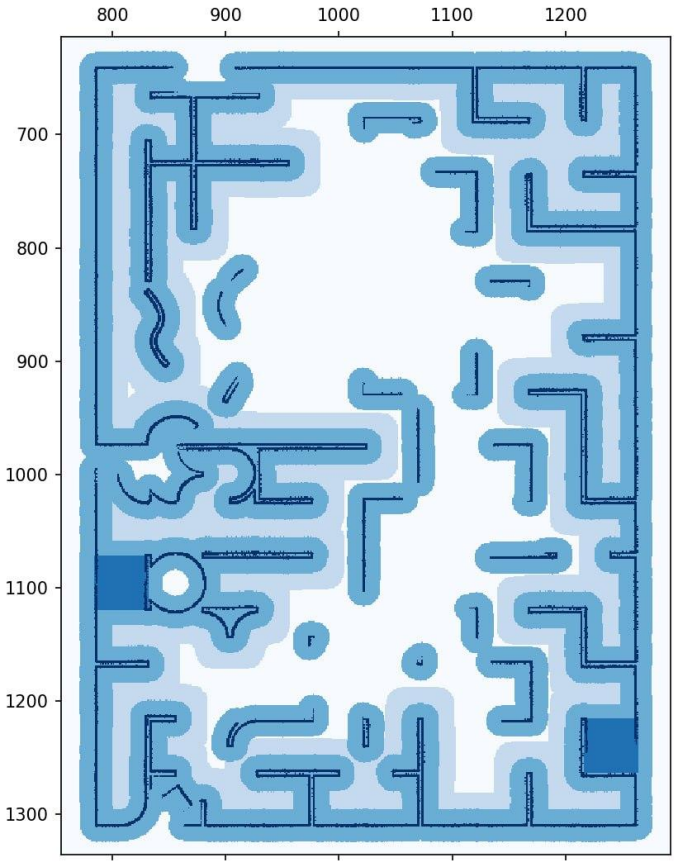
April 22, Alin, Michele

Tasks done today

- To systematically explore the entire environment, we implemented a visiting strategy that involved marking the empty spaces visited by the robot as "visited" on the map, as shown in figure 1. This information was then utilized to determine when to stop visiting a particular region and identify new unvisited regions adjacent to wall dilations. To discover these new spaces, we employed a breadth-first search (BFS) algorithm, which traversed the map and identified the unvisited areas for further exploration.

Figures/Drawings/Tables

Figure 1, the occupancy grid, highlighting the distinction between free space and visited space



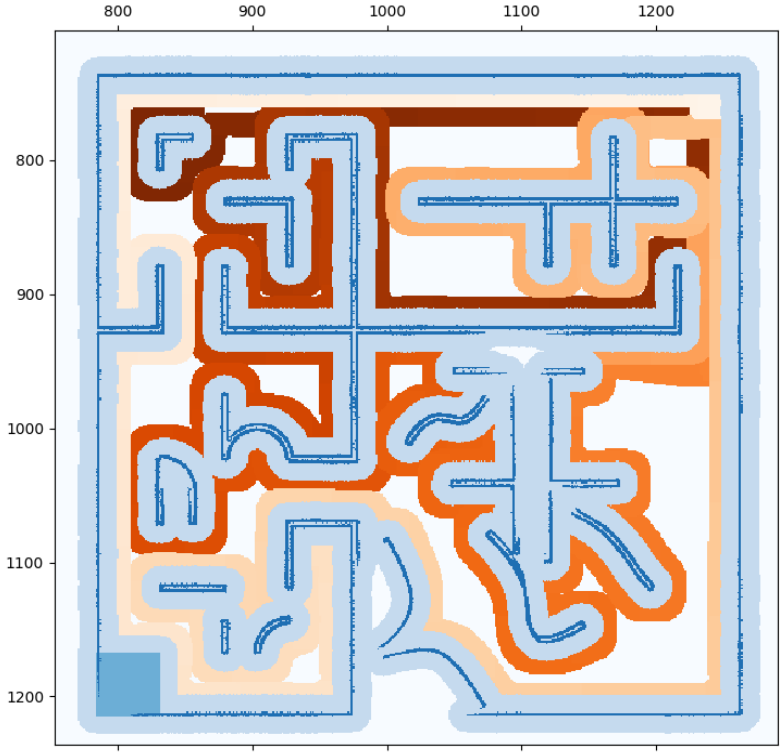
April 25, Alin, Michele

Tasks done today

- In response to the challenges encountered in path planning using the previous approach of marking regions as visited, we have implemented a revised method. Now, each visited region is assigned a unique number based on the time of exploration. This number is then utilized to determine whether to continue exploring or to stop visiting a region, even if it has been visited before. Figure 1 showcases the updated grid, reflecting this new approach.

Figures/Drawings/Tables

Figure 1, plot of the occupancy grid showing the visited areas based on the time of exploration



April 29, Alin, Michele

Tasks done today

- As our national competition approaches, we are currently fine-tuning and testing our robot.
- We have prepared an initial document that contains the information we intend to showcase on the poster.

May

Throughout the month of May, we conducted numerous experiments. However, we achieved limited results apart from the corrections made to the bonus map grid.

May 1, Alin, Michele

Tasks done today

- Incorporating a color sensor into our robot, we can now detect special tiles such as swamp and zone tiles. This sensor has enabled us to accurately identify and incorporate these tiles into the final bonus map grid.

Issues and solutions

We encountered an issue when detecting colored tiles along the borders, as our current snapping mechanism occasionally placed the tile in the wrong cell on the bonus map grid. To address this, we implemented a solution that takes into account the robot's distance from the tile border when adding a colored tile to the bonus map grid. By ensuring the robot is near the center of the tile, we eliminated the occurrence of random tile placements caused by snapping errors.

May 9, Alin, Michele

Tasks done today

By refining our occupancy grid to bonus grid conversion process, we have successfully eliminated certain non-feasible wall formations from the bonus map grid. This includes walls indicated in Figure 1, which are now excluded from the final representation. Figure 2 presents another bonus map grid incorporating our corrections.

Figures/Drawings/Tables

Figure 1, on the left the errors observed in the bonus map grid and on the right the corresponding occupancy grid

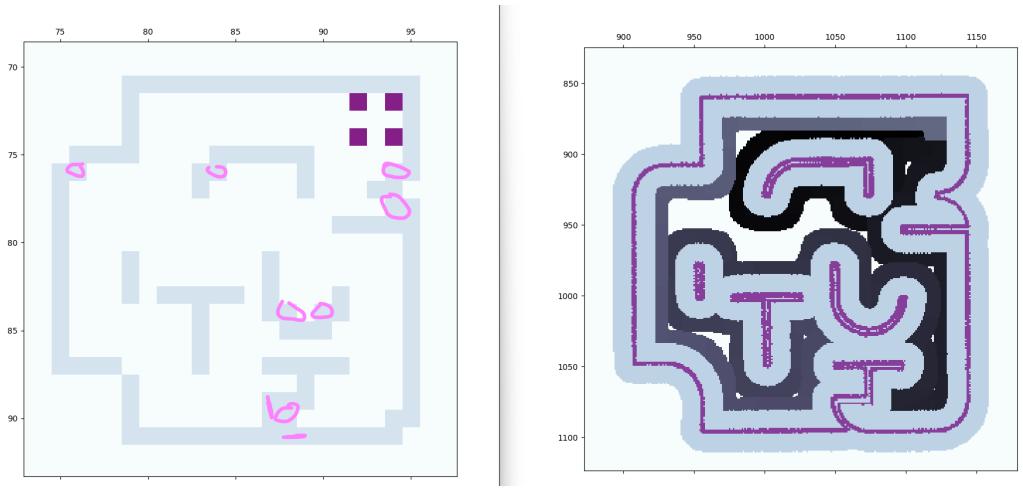
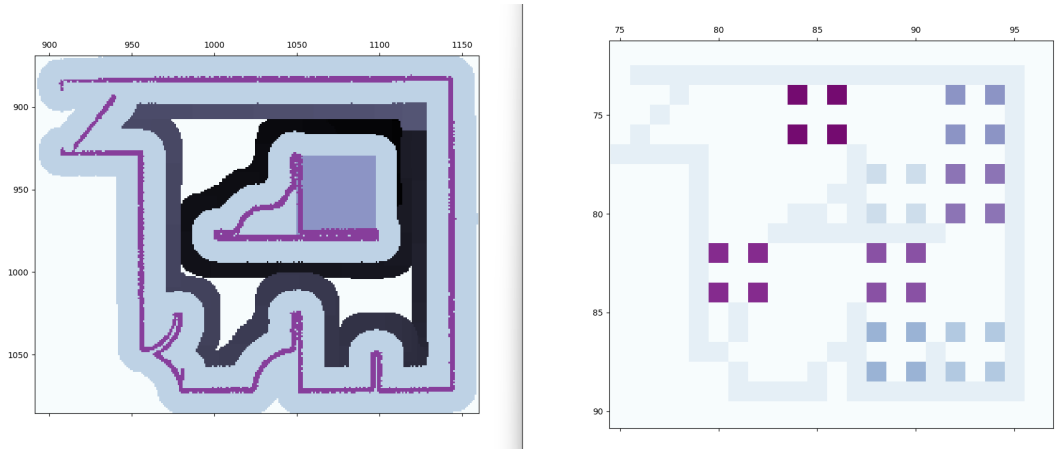


Figure 2, on the left the occupancy grid and on the right the corresponding bonus map grid with our new corrections



May 14, Alin, Michele

Tasks done today

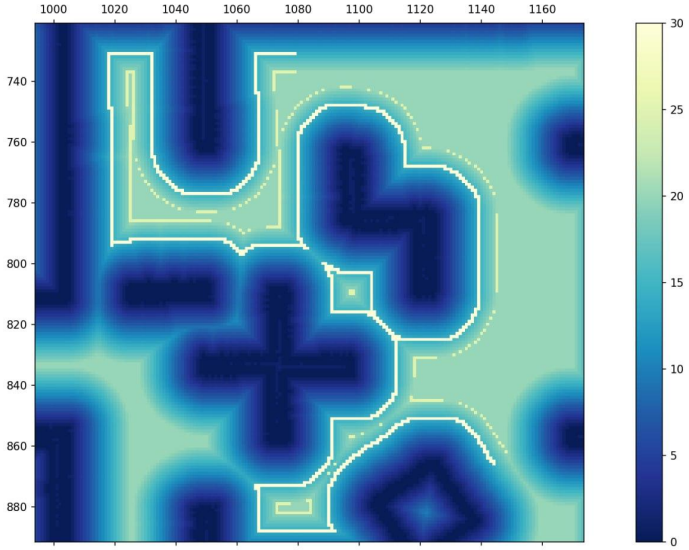
- Due to the robot's proximity to walls, we encountered difficulties in recognizing signs and navigating the maze efficiently. To address this, we devised a new path-planning strategy that follows the walls but maintains a certain distance from them whenever possible. We explored various approaches, one of which involved applying a Euclidean distance transform to the map to determine the minimum distances from the walls at each point. Subsequently, we implemented an algorithm to adjust the existing path, distancing it from the walls where feasible. It was important for us to avoid simply increasing the dilation radius, as this would have hindered navigation through narrow passages in the maze. Figure 1 showcases both the Euclidean transform and the path obtained from our previous algorithm, as well as the newly modified path.

Issues and solutions

The path we acquired deviates from our intended plan. Thus, we will endeavor to explore alternative solutions in the future.

Figures/Drawings/Tables

Figure 1, the old path and the new path we derived from our recent efforts



May 20, Alin, Michele

Tasks done today

- We have made the decision to relocate our cameras from the front to the sides of the robot. This adjustment is necessary as our current path planning occasionally causes the robot to overlook certain sections of the walls. Figure 1 shows the updated camera placements.
- We have resolved the previous path planning issue by employing a Laplace transform on the distance image. This transformation enabled us to identify a path that closely aligns with our desired outcome. Figure 2 displays the image with the Laplace transform applied, while Figure 3 showcases the path we achieved through processing the transformed image.

Figures/Drawings/Tables

Figure 1, the updated camera placements, with one positioned on each side

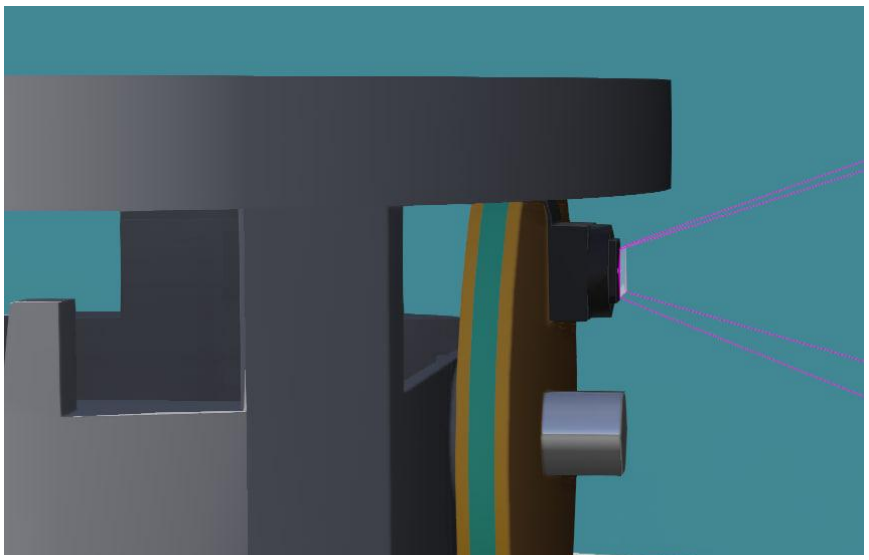


Figure 2, the image with the Laplace transform applied

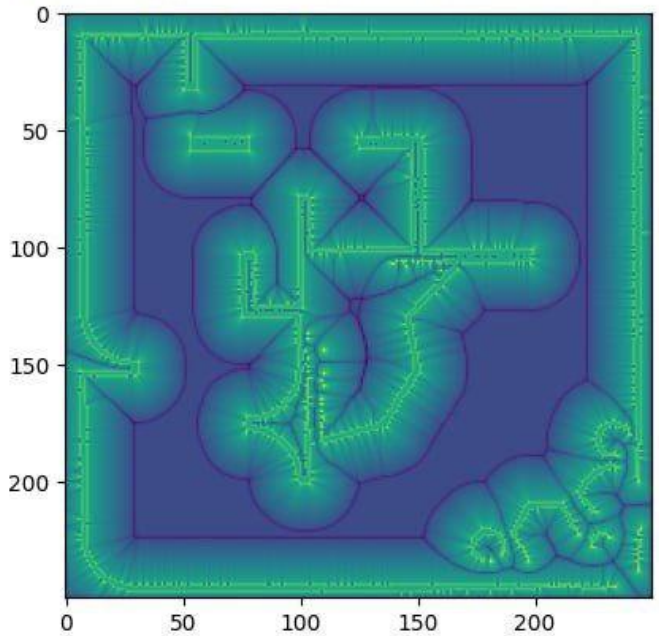
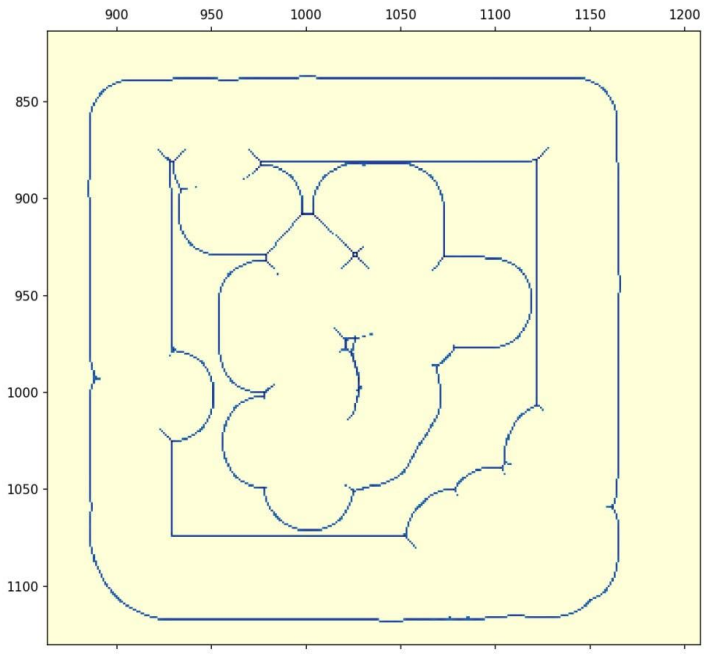


Figure 3, the path we achieved through processing the transformed image



June

In June, similar to May, we carried out a considerable number of experiments. As a result, we made significant decisions, such as transitioning to YOLO for object detection, replacing the previous model.

June 1, Alin, Michele

Tasks done today

- In our endeavor to enhance our system, we conducted experiments employing a probabilistic approach with a logit function on our occupancy grid. Regrettably, these efforts did not yield substantial improvements. As a result, we have decided to revert to our original non-probabilistic grid, continuing to place lidar hit points on the grid in the same manner as before.
- To enhance our model, we embarked on a transition from image classification to real-time object detection. After thorough research on options like RCNN and YOLO, we ultimately chose YOLO as our framework of choice. Subsequently, we obtained a functional implementation of YOLO and we started reconfiguring our dataset accordingly to align with this model.

Research

Logit function:

- https://en.wikipedia.org/wiki/Logit

YOLO:

- https://github.com/pjreddie/darknet
- https://github.com/WongKinYiu/yolov7
- https://arxiv.org/abs/2207.02696

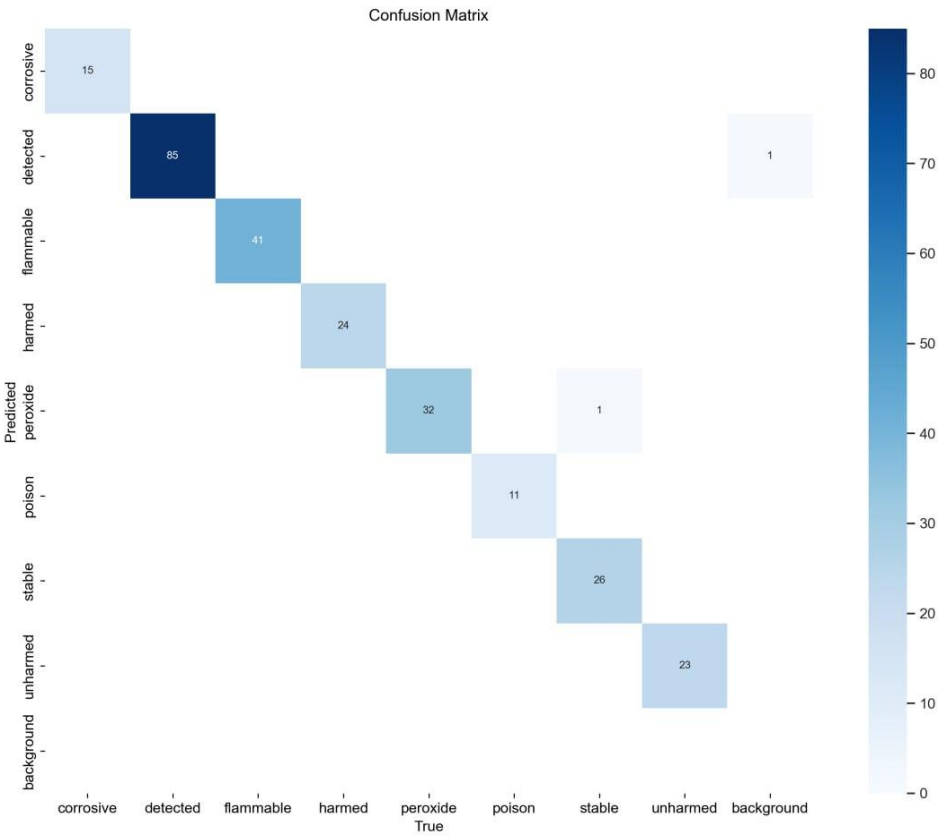
June 6, Alin, Michele

Tasks done today

- We successfully trained the YOLO implementation using our new dataset. Figure 1 presents the confusion matrix, providing an evaluation of our model's performance.

Figures/Drawings/Tables

Figure 1, confusion matrix, providing an evaluation of our model's performance



June 10, Alin, Michele

Tasks done today

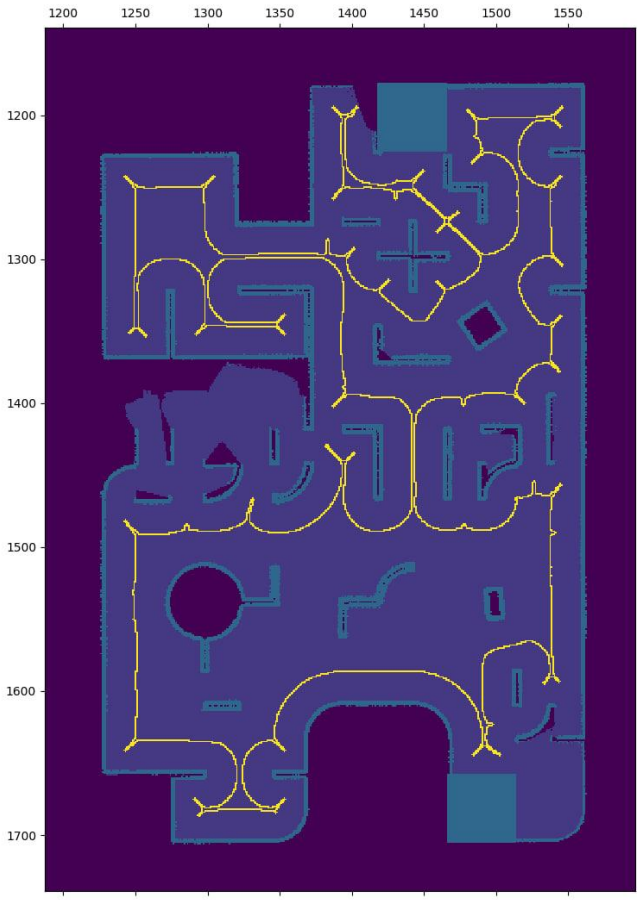
- After investing considerable effort into our new path planning algorithm, we have successfully computed a navigable path, as depicted in Figure 1. However, due to time constraints and the substantial additional work required to implement and utilize this approach, we have made the decision to abandon it. We have other commitments in our personal lives as well as other preparations for the upcoming competition that demand our attention.

New Ideas | Thoughts

With the limited time remaining, our plan is to enhance the dataset for our model, as it is still not perfect. Furthermore, we aim to incorporate the position of signs into the bonus map grid. Additionally, we intend to conduct thorough testing, address any issues with our current code, and implement improvements to enhance its functionality.

Figures/Drawings/Tables

Figure 1, the path obtained by processing the Laplace transformed grid while navigating the maze



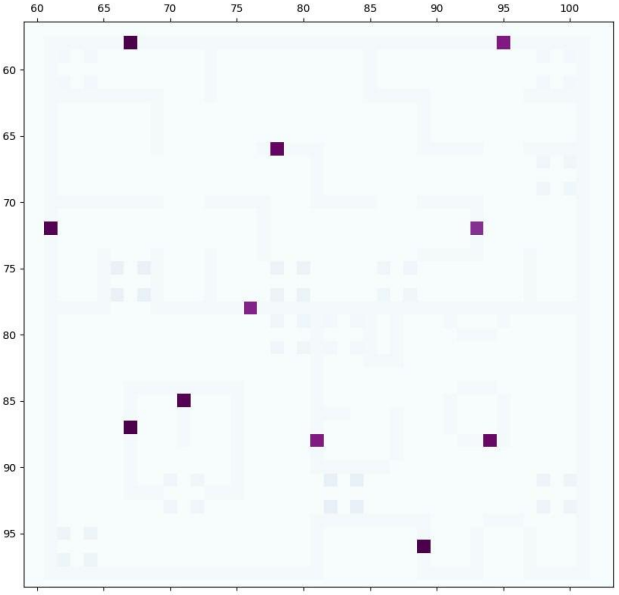
June 15, Alin, Michele

Tasks done today

- We have integrated sign placements into our bonus map grid. Figure 1 showcases the updated grid, featuring the newly added signs.

Figures/Drawings/Tables

Figure 1, bonus map grid with recently added signs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